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Information Systems Research

Publication details, including instructions for authors and subscription information:
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To cite this article:
Elina H. Hwang, Stephanie Lee (2025) A Nudge to Credible Information as a Countermeasure to Misinformation: Evidence from Twitter. Information Systems Research 36(1):621-636. https://doi.org/10.1287/isre.2021.0491

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INFORMATION SYSTEMS RESEARCH
Vol. 36, No. 1, March 2025, pp. 621–636
ISSN 1047-7047 (print), ISSN 1526-5536 (online)

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A Nudge to Credible Information as a Countermeasure to Misinformation: Evidence from Twitter

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Received: September 22, 2021
Revised: October 16, 2022; September 16, 2023; January 16, 2024
Accepted: January 31, 2024
Published Online in Articles in Advance: February 28, 2024

https://doi.org/10.1287/isre.2021.0491

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Abstract. Fueled by social media, health misinformation is spreading rapidly across online platforms. Myths, rumors, and false information on vaccines are flourishing, and the aftermath can be disastrous. A more concerning trend is that people are increasingly relying on social media to obtain healthcare information and tending to believe what they read on social media. Given the serious consequences of misinformation, this study aims to explore the efficacy of a potential cure for the infodemic we face. Specifically, we focus on a countermeasure that Twitter used, which is to nudge users toward credible information when users search topics for which erroneous information is rampant. This Twitter's policy is unique, in that the intervention is not about censorship but about redirecting users away from false information and toward facts. Our analysis uses 1,468 news articles that contain misinformation about health topics such as measles, vaccines, and cancer. Our analysis reveals that Twitter's nudging policy reduces misinformation diffusion. After the policy introduction, a news article that contains misinformation is less likely to start a diffusion process on Twitter. In addition, tweets that contain a link to misinformation articles are less likely to be retweeted, quoted, or replied to, which leads to a significant reduction in the aggregated number of tweets each misinformation article attracts. We further uncover that the observed reduction is driven by the decrease both in original tweet posts—those that first introduce misinformation news articles to the Twitter platform—and in those resharing the misinformation, although the reduction is more significant in resharing posts. Last, we find that the effect is driven primarily by a decrease in human-like accounts that share links to unverified claims but not by a decrease in activities by bot-like accounts. Our findings suggest that a misinformation policy that relies on a nudge to a credible source rather than on censorship can suppress misinformation diffusion.

History: Anandasivam Gopal, Senior Editor.
Supplemental Material: The online appendices are available at https://doi.org/10.1287/isre.2021.0491.

Keywords: misinformation • diffusion • online platform • countermeasure

1. Introduction

Fueled by social media, misinformation is spreading rapidly across the globe. By the term "misinformation," we refer collectively to unverified claims, false content, hoaxes, rumors, and conspiracy theories, following the industry convention (Wardle 2018). Misinformation about vaccines has led to undesirable social consequences. Although the measles vaccine—measles, mumps, and rubella (MMR)—is among the most-studied medical products, given safely to millions of children and adults each year (Jenco 2019), unverified claims about its side effects continue to spread online. A 1998 study alleged a connection between the measles vaccine and autism. Although it was later ruled that the claim lacked sufficient scientific evidence and the article was officially retracted from a journal, the myth surrounding the measles vaccine persists (Petrovic et al. 2001, Larson et al. 2011), leading to a rising number of children left unvaccinated (Evrony and Caplan 2017). As a result, we see a growing number of measles cases: According to the Centers for Disease Control and Prevention (CDC), the number of measles cases in the United States in 2019 hit its highest level in more than 25 years. The resurgence calls for unnecessary expenditures to "re"-combat a disease that was once declared eliminated in the United States (Centers for Disease Control and Prevention 2019).

Despite the substantial societal impact of misinformation, there is limited research evaluating the effectiveness of countermeasures implemented by social media platforms to combat online misinformation. To advance our understanding, this paper aims to empirically evaluate one of the countermeasures implemented by Twitter. In response to the increasing amount of

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misinformation spreading on its platform, Twitter initiated a misinformation policy in May 2019. This policy works in the following way. When a user searches for keywords such as vaccine or measles on its platform, Twitter triggers a prompt that contains the phrase "Know the Facts" on top, followed by the user's search results. The prompt also prints the following sentence: "To make sure you get the best information on vaccinations, resources are available from the U.S. Department of Health & Human Services," with a link to this trustworthy information source. The policy does not apply to all topics but only to specific topics such as vaccines and measles. Although misinformation also exists on other health topics, such as cancer or asthma, Twitter's intervention does not apply to them.

Twitter's policy is unique in that the intervention is not about censorship. A common approach to curbing misinformation has been to review posted content retrospectively and delete misinformation. For instance, Facebook uses a false news detection algorithm and human content moderators to censor posted misinformation. The drawback of this approach is that corrective action can only be taken after misinformation spreads and that it harms freedom of speech, which invites a heated debate around the topic of Internet censorship (Suciu 2019, Novacic 2020). In contrast, Twitter's nudging policy is not retrospective. Its focus is to help users find credible information for a topic where misinformation is rampant rather than to drop misinformation that users already posted.

This study empirically examines whether and, if so, how much Twitter's misinformation policy suppresses health-related misinformation diffusion. Although extant literature offers valuable insight into how misinformation propagates on online platforms (Stella et al. 2018, Vosoughi et al. 2018), to the best of our knowledge, few studies examined the efficacy of social media platform's policy in suppressing online misinformation. For our empirical analysis, we leverage a unique quasi-experiment setting where the treatment is the launch of Twitter's nudging policy. We evaluate variations in Twitter activity pre and post this policy rollout, distinguishing between misinformation affected by the policy and misinformation that is not. However, estimating a social media platform's policy is not straightforward. The transient nature of social media data, coupled with the inherent challenges of a quasi-experiment setting tied to the nonrandom selection of a treatment group, presents a set of challenges. Although we consciously discuss these intricacies and actively implement various steps to enhance the reliability of our analysis, we acknowledge that we may not be able to perfectly isolate the treatment effect. Our results show that a misinformation policy that relies on a nudge to a credible source rather than on censorship can play a crucial role in mitigating the spread of misinformation.

2. Conceptual Background

2.1. Related Literature

Our paper is related to several streams of literature. The most closely related works are those that empirically evaluate social platforms' interventions against misinformation. To the best of our knowledge, Chiou and Tucker (2018) is the only work that systematically examines the impact of an intervention actually used by social platforms. They studied whether Facebook's advertising ban—which prevents advertisers from running advertisements that link to fake news (Seetharaman 2016)—reduces the sharing of misinformation. Our study contributes to this scant body of literature by quantifying the effect of Twitter's prewarning-based intervention against misinformation.

Extant empirical studies provide rich descriptive analyses of the misinformation diffusion process. For instance, studies have compared diffusion patterns of false news versus true news. They documented that false news tends to spread faster, deeper, and farther than true news (Friggeri et al. 2014, Chiou and Tucker 2018, Vosoughi et al. 2018). From a theoretical perspective, several papers examined content-sharing patterns using an agent-based simulation model. For instance, Acemoglu et al. (2021) present an agent-based simulation model that studies how homophily may differentially influence high- and low-reliability content sharing online. Acemoglu and Gheibi (2010) and Acemoglu and Fagnani (2013) study how persuasive agents may facilitate the spread of misinformation and interfere with the convergence of beliefs among agents. Mostagir et al. (2022) and Mostagir and Siderius (2023) study how a social learning process might be manipulated by persuasive agents and characterize social networks that are more resistant to such manipulation. Our study complements this stream of literature by demonstrating how a prewarning countermeasure may alter misinformation diffusion patterns and how its impact might differ across different types of users.

Our study is also related to a stream of research that studies tactics to reduce misinformation diffusion. Tactics such as changing the wording of misinformation correction have been tested, but most were unsuccessful in strengthening the correction effect (Ecker et al. 2010). In addition, recent empirical studies explored potential interventions that online platforms may adopt to address the continued influence of misinformation. For instance, Kim et al. (2019) ran online experiments to understand whether the reputation rating systems of news sources may influence subjects' tendency to believe fake news. Similarly, Moravec et al. (2020) examined how different types of false news flags influence subjects' acceptance of fake news. Several theoretical papers also proposed platform designs that curtail misinformation diffusion. Candogan and Drakopoulos

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(2020) studied optimal signaling of content accuracy in social networks that maximizes engagement while minimizing misinformation diffusion. Our study contributes to this stream of research by studying the prewarning tactic as a way to reduce online misinformation diffusion.

2.2. Efficacy of Twitter's Misinformation Policy

Twitter's policy, although commendable for avoiding censorship, is uncertain in its effectiveness because the policy does not remove misinformation from Twitter. Various outcomes are possible. For instance, Twitter's policy may reduce misinformation diffusion by alerting users to upcoming misinformation. Prior psychology literature suggests that people are more successful in discounting invalid information when they are preemptively alerted that some of the information they are about to see may be misleading (Schul 1993, Chambers and Zaragoza 2001, Ecker et al. 2010). This can be attributed to a pre-exposure warning that mentally prepares individuals to be skeptical of upcoming information. This heightened skepticism enables users to scrutinize information more critically, making them less prone to accept false claims. Thus, Twitter's policy may prompt users to be cognitively more on guard, diminishing their likelihood to accept and further disseminate misinformation.

Nonetheless, a pre-exposure warning alone might not be sufficient to increase people's vigilance for upcoming misinformation. For instance, in a controlled experiment by Schul (1993), participants were presented with both invalid and valid information to make a judgment. The study revealed that a mere preliminary instruction to disregard invalid data did little to aid participants in distinguishing between valid and misleading information. Participants were better able to discount false information when reminded of the warning while they process information. Similarly, Ecker et al. (2010) found that vague warnings, merely hinting at the presence of misinformation without providing concrete examples, fall short in alerting individuals. Twitter's intervention is not accompanied by timely reminders nor concrete examples about people's continued reliance on misinformation. Rather, it offers a generalized statement, directing users to access reliable information via a provided link. Furthermore, this warning appears at the top of the screen, not immediately juxtaposed to the misleading content within a tweet. Hence, there remains a plausible concern that Twitter's policy might not achieve its intended objective of curtailing the spread of misinformation.

Additionally, even if we presume that Twitter's policy effectively deters individuals from posting or resharing misinformation, its overarching impact remains ambiguous due to several considerations. First, misinformation is spread not only by people but also by social bots. Defined as social media accounts largely or entirely steered by algorithms to generate and spread content (Yang et al. 2019), not all these bots have malicious intents. However, a significant portion has been identified as contributors to the misinformation diffusion (Shao et al. 2018). Because Twitter's policy is triggered by topic searches—a behavior exhibited by human users—it may not effectively curtail content disseminated by bots. Therefore, if social bots play an active role in spreading misinformation, the magnitude of the policy's impact in curbing misinformation spread may be small and vice versa. Second, not all users resort to searching before tweeting or retweeting. Because the policy is anchored around the search behavior, the proportion of users it actually impacts remains uncertain. Given that prior theory does not provide conclusive insights into the direction and magnitude of the policy's effect, there is a need for empirically exploring the effect of Twitter's policy. This study seeks to shed light on our understanding of how prewarning based countermeasure influences misinformation diffusion on social platforms.

3. Empirical Setting and Data

3.1. Twitter's Misinformation Policy

On May 10, 2019, Twitter announced the initiation of a misinformation policy to protect the health of public conversation (Harvey 2019). The misinformation policy involves the following intervention. When a user searches for certain keywords associated with a topic about which misinformation is abundant, a prompt with a link to a credible information source appears on top of the screen, followed by tweets that include the search term. Twitter introduced this policy to combat a growing amount of false information about measles and vaccines. Because the misinformation policy applies to certain topics (i.e., vaccine, measles) but not all, it provides a quasi-experiment setting where we can quantify its effectiveness in curbing misinformation diffusion. Figure 1 presents screenshots of the prompt triggered by the policy when we searched for the keyword 'vaccines' and 'measles,' respectively. In response to these keywords, Twitter nudged us to a trustworthy information source, 'vaccine.gov' so that we can obtain credible information about measles vaccines.

3.2. Data

For our empirical investigation, we need to track the spread of misinformation on Twitter. To this end, we need the following data: (1) news articles containing misinformation ("misinformation articles") and (2) Twitter activities involving these articles.

We first collected misinformation articles about health topics using the Hoaxy platform.[1] The Hoaxy platform

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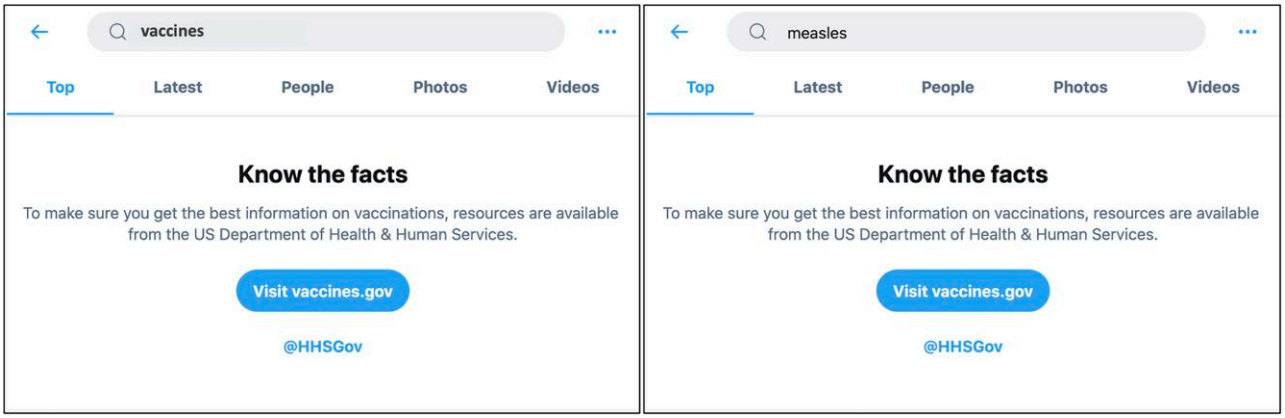
Figure 1. (Color online) Intervention by Twitter's Misinformation Policy



was developed by the Network Science Institute at Indiana University to track and collect online misinformation. The Hoaxy platform does not perform fact-checking by itself. Instead, it retrieves news articles that have been determined to contain misinformation. The Hoaxy platform collects a very comprehensive set of misinformation articles by tracking news articles published by the following two types of websites: (1) misinformation articles from independent fact-checking organizations such as FactCheck.org, PolitiFact.com, and Snopes.com and (2) news articles from low-credibility news sources. The low-credibility news sources include over 800 websites that frequently publish misinformation articles. The domains of the low-credibility news sources are identified and continuously updated by fact-checking organizations (Shao et al. 2016). Examples of low-credibility news sources include BuzzFeed fake news sites,[2] FactCheck.org misinformation directory,[3] PolitiFact fake news almanac,[4] and Wikipedia list of fake news websites.[5] To automatically gather articles containing misinformation from these news and fact-checking sites, the Hoaxy platform uses a system of crawlers and databases (Shao et al. 2016).

Using Hoaxy API,[6] we identified a total of 1,468 articles containing misinformation about several health topics (e.g., measles, vaccine, cancer, asthma), published over a period from May 2016 to October 2019. To ensure that the collected articles contain misinformation about health topics, we manually screened all misinformation articles collected from the Hoaxy API. Among the 1,468 articles we identified, 855 articles (58.24% of all) contain misinformation about vaccines or measles and were subject to Twitter's misinformation policy. The remaining articles include misinformation about some other health topics not subject to the misinformation policy (e.g., cancer, asthma). For each article, we collected its title, the date published, the publisher name, and the article's URL. Table 1 presents samples of the misinformation articles we collected.

Second, to capture how misinformation spreads on Twitter, we collected all public tweets that include URL links to the 1,468 misinformation articles. For this data collection, we used both Hoaxy API and Twitter API.[7] Because tweets are limited to only 140 characters, most news articles are shared by including URLs to the articles. To track how articles are shared on Twitter, Hoaxy API uses a filter endpoint of Twitter API to collect tweets containing URLs of articles. Because URLs of the same web page may have different forms based on different protocol schema and query parameters, Hoaxy API relies on canonical URLs. Some tweets contain t.co links, which is a shortened link by Twitter. Twitter API provides an expanded URL that t.co links redirect to. Therefore, Hoaxy API uses an expanded URL field to identify canonical URLs of news articles.

We found that 249 of 1,468 misinformation articles (≈17% of all) did not start the diffusion process on Twitter (i.e., there are zero tweets with the link to the

Table 1. Misinformation Articles

Title	Published
Measles vaccines kill more people than measles, CDC data proves	July 2016
Scientists find root that kills 98% of cancer cells in only 48 hours	September 2016
How cannabis treats asthma better than any drugs	February 2017
Aluminum in vaccines may cause autism according to experts in new study	December 2017
Bill Gate's former doctor says billionaire refused to vaccinate his kids	February 2018
Science CONFIRMS it: Vaccines spread measles	January 2019

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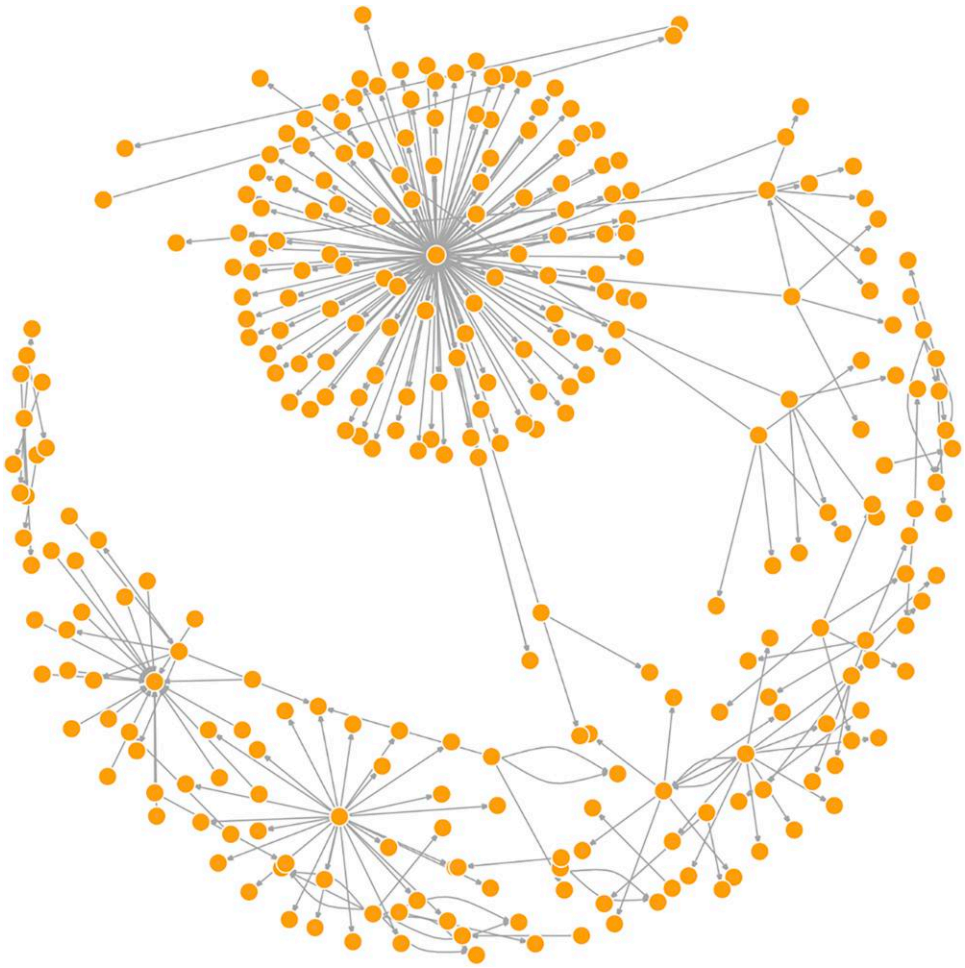
Figure 2. (Color online) Misinformation Diffusion Network



Notes. A diffusion network of the article titled, "Measles vaccines kill more people than measles, CDC data proves." Published in July 2016 and tweeted 518 times.

articles). For the remaining 1,219 misinformation articles (≈83% of all), we collected all 35,578 public tweets posted during our empirical period. Because Hoaxy API does not provide detailed individual tweet characteristics, we used Twitter API to complement our data with individual tweet characteristics. For each tweet, we collected its unique tweet ID, date/time posted, tweet text, and whether it was reshared by another user. If a tweet was reshared, we also saved information on how it was reshared (i.e., retweeted, replied, or quoted). Although many misinformation articles received multiple tweets, some were tweeted only once: One user posted an original tweet linking to the article, but it was not reshared by other users afterward. On average, those misinformation articles that started a diffusion cascade in Twitter were tweeted around 29 times. A few articles were tweeted to a substantial degree, reaching up to the maximum tweet count of 2,073. In terms of tweet types, original tweets (tweets that first introduce a misinformation article) and retweets comprise the majority of the tweets. The 35,578 tweets consist of 16,139 original tweets, 16,526 retweets, 1,713 quotes, and 1,200 replies. Figure 2 illustrates a diffusion network of one misinformation article in our data set: "Measles vaccines kill more people than measles, CDC data proves," which was tweeted 518 times.

In addition, we collected several user characteristics of each tweet poster using Twitter API, to gain a nuanced understanding of how the impact of the misinformation policy varies across different user characteristics. For each Twitter user, we collected user I.D., username, the number of followers (followers_count), the number of other users the focal user is following (friends_count), the number of tweets the focal user has liked in the account's lifetime (favourites_count), the number of public lists that the focal user is a member of (listed_count), the number of tweets and retweets issued by the focal user (statuses_count), and the Twitter join date. We also collected the likelihood of a Twitter user account being a social bot (user botscore) to understand

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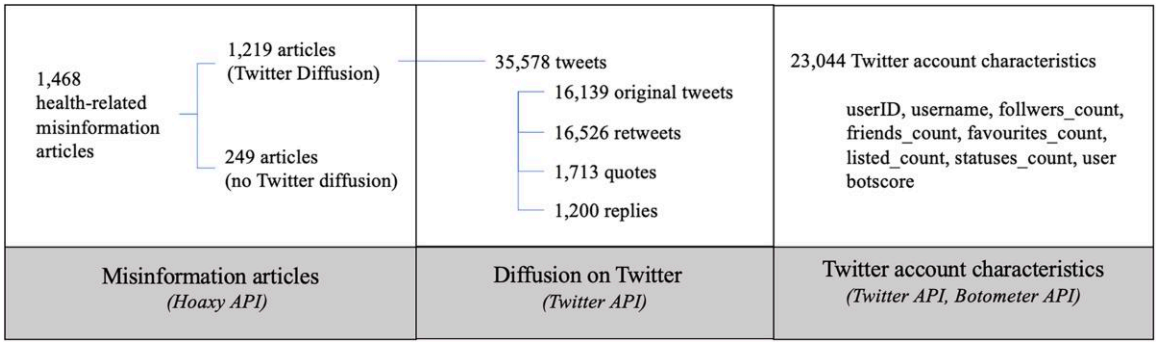
Figure 3. (Color online) Data Collection Process



the heterogeneous effect of Twitter's policy on misinformation diffusion by human-like and bot-like Twitter accounts. We present more details about the user botscore in Section 6.2. Figure 3 summarizes our data collection process.

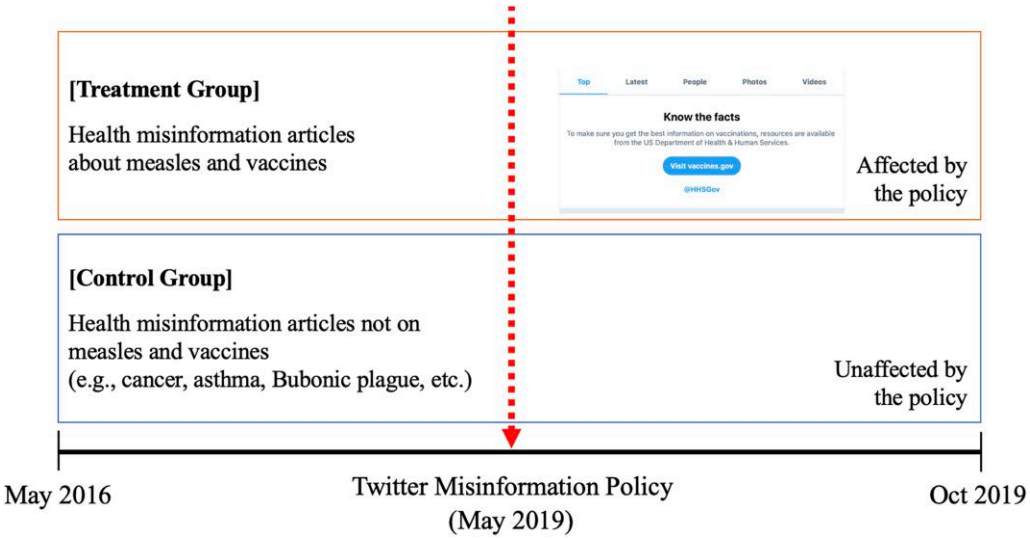
4. Econometric Model

4.1. Quasi-Experiment

We leverage a quasi-experiment setting to estimate the impact of the Twitter policy on misinformation spread. Figure 4 depicts our quasi-experimental setup. In our empirical setting, the treatment is the implementation of the Twitter policy, and Twitter users are exogenously treated as the users could not have forward-looking expectations of the policy change. As mentioned in Section 3.2, among the 1,468 misinformation articles we collected, 1,219 articles contain misinformation about measles or vaccines, which are subject to Twitter's misinformation policy. The remaining articles include misinformation about other health topics that are not subject to the misinformation policy. Consequently, the 1,219 articles about vaccines or measles form the treatment group and the remaining articles serve as a control group in our analysis.

Like other social media posts, tweet posts are, by nature, short lived. According to industry reports, the average lifespan of a tweet post is approximately 24 minutes (Graffius 2022), meaning it takes an average of 24 minutes for a tweet post to be buried in the news feed. In our empirical context, the median number of days an average misinformation article takes from the first tweet's appearance to the last tweet is two days. Because of the short lifespan of tweet activities of misinformation articles, it is not feasible for us to track many misinformation articles before and after the treatment: Only a few misinformation articles are actively going through the diffusion process before and after the treatment. Consequently, we use repeated cross-sectional data for estimation. Repeated cross-sectional data, rather

Figure 4. (Color online) Quasi-Experiment Setup



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than panel data, are often used when each observation covers a short time span (Baltagi 2008), and the number of time periods for each observation is limited (Hill et al. 2020).[8]

Different from panel data that follow the outcome of the same units across time, repeated cross-sectional data capture the outcome from different units across time. In our setting, we measure tweet activities of different sets of misinformation articles for different periods, some published before and others after the treatment, while ensuring comparability of articles in different periods. Specifically, based on the timestamp of individual tweets in our data set, we capture pretreatment tweet activities of articles published before the treatment. Similarly, we capture posttreatment tweet activities of articles published after the treatment.

4.2. Model

To examine the impact of the Twitter policy on the spread of health misinformation, we first examine whether the policy lowers the likelihood that a misinformation article starts the diffusion process on Twitter. We use the following conditional logit model to estimate the impact:

$$\begin{aligned} &TwitterDiffusion_{ijt} \\ &= \alpha + \beta_1 TreatedArticle_j + \beta_2 Post_t \\ &\quad + \beta_3 (TreatedArticle_j \cdot Post_t) + \boldsymbol{X}_{ijt} \cdot \delta + \varepsilon_{ijt}, \end{aligned} \tag{1}$$

where the outcome variable $TweetDiffusion_{ijt}$ captures whether misinformation article i in group j published in time t ever starts a diffusion process on Twitter. $TweetDiffusion_{ijt}$ equals one if there is at least one tweet that provides a link to the misinformation article, and zero otherwise. The variable $TreatedArticle_j$ equals one for misinformation articles that are affected by the Twitter policy and zero for the articles not subject to the Twitter policy. During our empirical window, Twitter applied the misinformation policy to keywords related to vaccines and measles. Hence, articles covering the topics of vaccines and measles belong to the treatment group ($j = 1$). Other articles containing misinformation about other health topics belong to the control group ($j = 0$); β_1 captures differences in tweet activities between treatment and control groups that could be the result of permanent differences between the two groups. That is, treated misinformation articles on vaccines and measles may be intrinsically different from other health-related misinformation articles (e.g., misinformation on cancer, asthma), which may lead to different patterns of tweet activities.

$Post_t$ is an indicator variable that equals one if article publication time t is after the implementation of Twitter's misinformation policy and zero otherwise. $(TreatedArticle_j \cdot Post_t)$ is an interaction of the variable $TreatedArticle_j$ and the variable $Post_t$, which indicates the treatment status. Articles are treated if the article covers the topics of vaccines or measles and if it is published after the policy implementation. Hence, $(TreatedArticle_j \cdot Post_t)$ equals one when $TreatedArticle_j = Post_t = 1$ and zero otherwise. In Equation (1), $\beta_3 < 0$ would indicate that the introduction of policy helps curb the spread of misinformation.

Next, to control for potential heterogeneities across news articles, we incorporate several covariates in our empirical model. Specifically, the vector term $\boldsymbol{X}_{ijt}$ includes $Sentiment_{ijt}$, $Subjectivity_{ijt}$, and $Length_{ijt}$.[9] The three variables control for heterogeneity of misinformation articles in terms of their title sentiment, subjectivity, and length. Sentiment measures the sentiment of the text; -1 indicates the most negative sentiment and 1 indicates the most positive sentiment. Subjectivity measures the amount of personal opinion included in the text, where 0 indicates the most objective text and 1 indicates the most subjective text. Length measures the number of words in the article title. In addition, the vector term $\boldsymbol{X}_{ijt}$ includes topic fixed effects, article publication month and year fixed effects, and media source fixed effects. Topic fixed effects tease out the mean differences of Twitter activities across articles covering different topics. For example, misinformation articles covering vaccines may attract a significantly broader audience than others covering cancer. With the inclusion of the topic effects, the variable $TreatedArticle_j$ is absorbed by the fixed effects. The article publication month-year fixed effects control for shocks related to time that are common across news articles. For example, articles that are published earlier would have a longer time to attract more Twitter activities. Also, people may have different propensities to use Twitter across different seasons. The article publication month-year fixed effects capture such heterogeneities related to time. The media source fixed effect teases the mean differences in Twitter activities across articles published by different media. Some media may have a broader audience than others, and some may have a better reputation than others, which may influence Twitter activities its article attracts. ε_{ijt} is an idiosyncratic random error term.

In addition, conditional on the Twitter diffusion started, we further examine the effect of the Twitter policy on the total number of tweets of treated misinformation articles (i.e., treated group) and untreated misinformation articles (i.e., control group) before and after the Twitter policy implementation. For this estimation, we replace the dependent variable in Equation (1) to $Tweet_{ijt}$ and use a Poisson model. The outcome variable $Tweet_{ijt}$ captures the total Twitter activities involving article i in group j published in time t, which is operationalized as the total number of tweets that include the link to the article. The Poisson model is

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Table 2. Correlations of Model Variables

	Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1)	$TwitterDiffusion_{ijt}$	1.000								
(2)	$Tweet_{ijt}$	0.116***	1.000							
(3)	$OriginalTweet_{ijt}$	0.175***	0.789***	1.000						
(4)	$NonOriginalTweet_{ijt}$	0.081**	0.972***	0.622***	1.000					
(5)	$ReTweet_{ijt}$	0.076**	0.964***	0.603***	0.997***	1.000				
(6)	$Quote_{ijt}$	0.080**	0.905***	0.620***	0.916***	0.898***	1.000			
(7)	$Reply_{ijt}$	0.077**	0.358***	0.359***	0.319***	0.254***	0.287***	1.000		
(8)	$TreatedArticle_j$	0.037	0.060*	0.068**	0.050‡	0.046‡	0.053*	0.058*	1.000	
(9)	$Post_t$	−0.008	0.032	0.046‡	0.023	0.025	0.017	−0.007	0.030	1.000

‡$p < 0.10$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

used as it is an appropriate count data model that is not subject to the incidental parameter problem (Greene 2002).[10] Table 2 provides correlations of the model variables. Table 3 summarizes the definition of all model variables, including the ones we use for the heterogeneous effect analysis in Section 6.

4.3. Identification Challenges

Our quasi-experiment setting and observational data create a few challenges in identifying effects. First, we use repeated cross-sectional data for estimation due to social media post's short lifespan. Although repeated cross-sectional data reduce concern about data's limited time periods, it requires more care to ensure data comparability. Different from panel data that capture outcomes of the same units across time, repeated cross-sectional data capture the outcome from different units across time. Hence, if the different units are not comparable, identification can be challenging. Second, we leverage a quasi-experiment setting for causal inference. Unlike randomized experiments, quasi-experiments aim to estimate an intervention's causal impact without random assignment of treatments. Because treatment may be assigned based on strategic reasons, quasi-experiments are subject to internal validity concerns (Cameron and Trivedi 2005). In our setting, Twitter strategically chose to apply its "Know the Facts" policy to the topics of vaccines and measles. Thus, systematic differences could exist across misinformation articles in treatment and control groups, and such differences may complicate the identification of effects.

We take multiple steps to strengthen our identification by enhancing the comparability of articles in different periods and groups. First, our model incorporates several fixed effects: (1) publication month-year fixed effect, (2) topic fixed effect, and (3) media source fixed effect. The fixed effects effectively tease out time-invariant mean differences across misinformation articles driven by a particular time, topic, and media source. Second, we include covariates capturing sentiment, subjectivity, and length of misinformation articles to control for heterogeneities of misinformation articles not captured by the fixed effects. Third, we check the robustness of our results with several variations of models and data: a model including topic-specific time trends (Online Appendix C), Heckman selection model

Table 3. Variable Definition

Variable name	Definition
$TwitterDiffusion_{ijt}$	An indicator variable which takes value of 1 if there is at least one tweet that provides the link to misinformation article i in group j published in time t, or 0 otherwise.
$Tweet_{ijt}$	The total number of tweets received by the misinformation article, conditional on the Twitter diffusion started for the article.
$OriginalTweet_{ijt}$	The total number of tweets that introduce the link of misinformation article to Twitter. This measure is analogous to the number of network cascades.
$NonOriginalTweet_{ijt}$	The total number of tweets that reshare existing tweets with the link to misinformation article. This number equals the sum of the total number of retweets, quotes, and replies with the link to misinformation article i.
$ReTweet_{ijt}$	The total number of retweets that provide the link to misinformation article.
$Quote_{ijt}$	The total number of quotes that provide the link to misinformation article.
$Reply_{ijt}$	The total number of replies that provide the link to misinformation article.
$TreatedArticle_j$	An indicator variable which takes value of 1 if misinfomration article is subject to the Twitter policy, or 0 otherwise.
$Post_t$	An indicator variable which takes value of 1 if article publication date t is after the implementation of Twitter's policy on May 10, 2019, or 0 otherwise.

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(Online Appendix D), matched data (Online Appendix E), constrained time window for Twitter activities (Online Appendix F), panel data analysis (Online Appendix G), and removal of observations that may be affected by another Twitter intervention (Online Appendix H). Also, like other quasi-experiment studies, we perform analyses to check whether the parallel trends assumption is viable in our setting: the relative time model (Online Appendix A) and a random implementation test (Online Appendix B). Last, we conducted a controlled experiment to further alleviate the concern about endogeneity and internal validity (Online Appendix I).

5. Results

Table 4 presents the results that explore the treatment effects of Twitter's misinformation policy on misinformation spread. Columns (1) and (2) in Table 4 incorporate month-year and topic fixed effects, and the variable $TreatedArticle_j$ is absorbed by the topic fixed effects. Columns (3) and (4) additionally include media source fixed effects, and columns (5) and (6) include additional covariates. Because the fixed effects and additional controls enhance identification, we use models with a full set of fixed effects and controls to interpret the results throughout this paper.

When the outcome variable measures $TwitterDiffusion_{ijt}$, the coefficient of the key variable $TreatedArticle_j \times Post_t$ is negative and statistically significant ($\beta = -1.749$, $p < 0.10$) (column (5)). Conditional on starting a Twitter diffusion, the Twitter policy reduces the number of tweets that misinformation articles receive. The Poisson model shows that the estimated coefficient is negative and statistically significant ($\beta = -1.967, p < 0.001$), which indicates a significant reduction in the number of tweets after the policy change (column (6)). We find that articles that are treated by the misinformation policy are less likely to be shared and disseminated than those that are not treated after the policy implementation, which suggests that the platform's misinformation policy can help curb the spread of misinformation.

6. Heterogeneous Effects

In this section, we examine the heterogeneous effect of the Twitter policy across different tweet types and Twitter users to better understand the sources of the reduction.

6.1. Effects Across Different Tweet Types: Original vs. Nonoriginal Tweets

A Twitter user can post a new original tweet or interact with an existing tweet via retweets, quotes, and replies. We classify each tweet as an *original* tweet or a *nonoriginal* tweet, where an original tweet refers to a tweet that introduces a misinformation link from a media source to Twitter and a nonoriginal tweet refers to a tweet that retweets, quotes, or replies to an existing tweet. For each misinformation article, we calculate the total number of original tweets and construct a variable, $OriginalTweet_{ijt}$. The variable is analogous to the number of a network's cascades, as introduced in Vosoughi et al. (2018). Additionally, we construct a variable $NonOriginalTweet_{ijt}$, which refers to the total number of nonoriginal tweets for each misinformation article.

Figure 5 illustrates a Twitter diffusion process of three hypothetical misinformation articles. In this illustration, articles A and C start diffusion on Twitter, whereas article B does not. For articles A and C, the

Table 4. Effects of Misinformation Policy on Misinformation Spread

Model	(1) Conditional Logit	(2) Poisson	(3) Conditional Logit	(4) Poisson	(5) Conditional Logit	(6) Poisson
Dependent variable	$TwitterDiffusion_{ijt}$	$Tweet_{ijt}$	$TwitterDiffusion_{ijt}$	$Tweet_{ijt}$	$TwitterDiffusion_{ijt}$	$Tweet_{ijt}$
$TreatedArticle_j \times Post_t$	−1.078‡	−1.807***	−1.800‡	−1.951***	−1.749‡	−1.967***
	(0.582)	(0.390)	(0.931)	(0.430)	(0.929)	(0.403)
$Post_t$	−0.0715	0.838	−2.948**	1.007*	−2.981*	1.028*
	(0.807)	(0.532)	(1.018)	(0.502)	(1.219)	(0.455)
$Sentiment_{ijt}$					−0.718	−0.341
					(0.603)	(0.264)
$Subjectivity_{ijt}$					0.0354	−0.0164
					(0.484)	(0.191)
$Length_{ijt}$					0.00595	0.0522*
					(0.102)	(0.0248)
Month-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Topic fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Media source fixed effects	No	No	Yes	Yes	Yes	Yes
Log likelihood	−506.66	−42,369.02	−218.04	−29,912.52	−217.09	−29,339.40
N	1,468	1,219	1,468	1,219	1,468	1,219

‡$p < 0.10$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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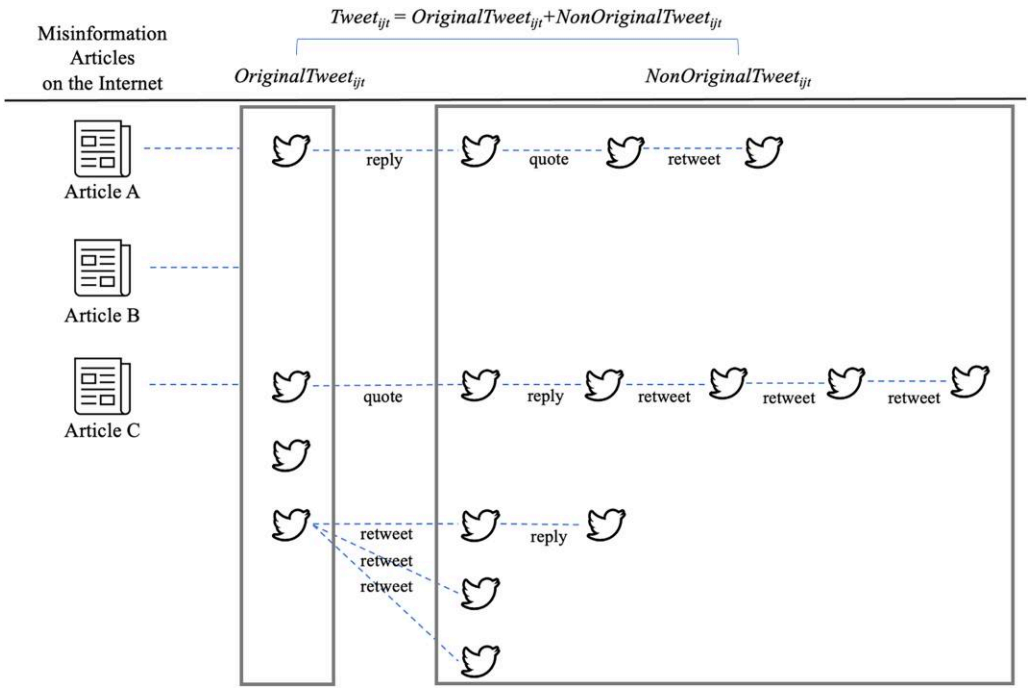
Figure 5. (Color online) Illustration of Misinformation Diffusion on Twitter



numbers of original tweets equal one and three, respectively. The number of original tweets for article B is zero. The numbers of nonoriginal tweets for articles A and C are three and nine, respectively. One of the dependent variables in our main analyses, $Tweet_{ijt}$, is the sum of original and nonoriginal tweets. By decomposing the total number of tweets each misinformation article garners from Twitter into original and nonoriginal tweets, we can better understand which activities Twitter policy discourages more: introducing misinformation to Twitter or resharing misinformation that is already on Twitter.

We use Equation (1) for this estimation, except that we use $OriginalTweet_{ijt}$ and $NonOriginalTweet_{ijt}$ as dependent variables. The results are presented in Table 5. All models in Table 5 incorporate month-year fixed effects, topic fixed effects, and media source fixed effect. We find that the Twitter policy reduces the number of both original and nonoriginal tweets that share links to misinformation articles, and a greater reduction is seen for nonoriginal tweets (Table 5, columns (1) and (2)). This result indicates that Twitter's intervention discourages people from both introducing and resharing misinformation on Twitter, while it discourages resharing more than introducing.

Columns (3)–(5) further show the decomposition of $NonOriginalTweet_{ijt}$ into the number of retweets ($ReTweet_{ijt}$), quotes ($Quote_{ijt}$), and replies ($Reply_{ijt}$). The results indicate that the policy reduces the reply tweets more than the other two types of resharing tweets

Table 5. Effects on Different Tweet Types: Original vs. Nonoriginal Tweets

Model Dependent variable	(1) Poisson $OriginalTweet_{ijt}$	(2) Poisson $NonOriginalTweet_{ijt}$	(3) Poisson $ReTweet_{ijt}$	(4) Poisson $Quote_{ijt}$	(5) Poisson $Reply_{ijt}$
$TreatedArticle_j \times Post_t$	−1.458***	−2.620***	−2.584***	−2.864***	−4.341***
	(0.438)	(0.367)	(0.392)	(0.431)	(0.724)
$Post_t$	0.519	1.515**	1.591**	1.154**	−0.967
	(0.392)	(0.479)	(0.488)	(0.408)	(1.496)
Controls	Yes	Yes	Yes	Yes	Yes
Month-year fixed effects	Yes	Yes	Yes	Yes	Yes
Topic fixed effects	Yes	Yes	Yes	Yes	Yes
Media source fixed effects	Yes	Yes	Yes	Yes	Yes
Log likelihood	−11,547.80	−20,861.45	−18,348.94	−2,382.83	−1,405.74
N	1,219	1,219	1,219	1,219	1,219

‡$p < 0.10$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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(retweets and quotes). A reply tweet is used to respond to another person's tweet directly and is more conversational than other types of resharing tweets. Consequently, a reply tweet is more frequently used among human accounts than bot accounts (Shao et al. 2018). Hence, this result might indicate that Twitter policy reduces misinformation diffusion through human accounts more than through bot-like accounts. In Section 6.2, we delve deeper into the heterogeneous effect of Twitter policy on the activities of human versus bot accounts.

6.2. Effects Across Different Twitter Accounts

6.2.1. Human vs. Bot Accounts. In this section, we further examine how Twitter's misinformation policy affects misinformation diffusion by human and bot Twitter accounts. A bot account refers to a social bot, which is a computer algorithm that automatically produces and shares content and interacts with humans on social media (Ferrara et al. 2016). Previous studies have developed machine-learning algorithms that assign a Twitter account the likelihood of its being a social bot (bot score). We use the algorithm developed by Varol et al. (2017) to collect bot scores of Twitter accounts in our data set. The bot score is computed using a classifier that evaluates the extent to which an account exhibits similarity to the characteristics of social bots. The classifier is based on a supervised machine learning algorithm leveraging more than 1,000 features[11] extracted from public data and metadata about Twitter accounts. Using the classifier, we assigned each Twitter account a bot score between zero and one; a higher bot score indicates that the Twitter account is likely to be a bot.

Based on the bot scores, we first calculate the total number of tweet activities generated by human-like accounts and bot-like accounts. Following the prior literature (Yang et al. 2019), a tweet activity is defined as bot-generated when $0.9 \leq$ Twitter account botscore ≤ 1 and defined as human-generated when $0 \leq$ Twitter account botscore ≤ 0.1. We find that, in our setting of health topics, tweets by bot-like accounts comprise a small portion (≈0.65% of all tweet activities in the data). Our results indicate that the misinformation policy does reduce tweets by human-like accounts (Table 6, column (1)), but does not have statistically significant effects on tweets by bots (Table 6, column (2)). Our analysis reveals that the reduction in the number of tweets that share links to fake news articles is driven primarily by the reduction in human users who share links to misinformation articles (column (1)). Given that Twitter's policy is triggered by searching for a disputable topic—which humans do, it makes sense that Twitter's policy does not affect the activities of social bots.

6.2.2. Popular vs. Less-Popular Accounts. We also examine heterogeneity in the effect of the Twitter policy across popular and less popular accounts. On Twitter, users can follow others' accounts by subscribing and then seeing these tweets on their timelines. Therefore, the number of followers indicates the popularity of an account. To analyze the differing effect of Twitter policy across the popularity of accounts, we compared the total number of tweet activities generated by popular versus less popular accounts. For this analysis, we define popular accounts as those whose number of followers is in the top 10% in our data and less popular accounts as the rest. We then run the analysis for each subsample and find that the policy significantly reduces tweets by both popular accounts (Table 6, column (3)) and less popular accounts (Table 6, column (4)), but we see a greater reduction in tweets by popular accounts.

The finding that the Twitter policy suppresses misinformation diffusion by popular accounts is promising. The seminal work by Katz and Lazarsfeld (1955) describes the essential role of opinion leaders in the

Table 6. Heterogeneous Effects of Misinformation Policy on Different Users

	(1)	(2)	(3)	(4)	(5)	(6)
Model	Poisson	Poisson	Poisson	Poisson	Poisson	Poisson
Dependent variable	*Human*	*Bot*	*Popular*	*Less Popular*	*Active*	*Less Active*
$TreatedArticle_j \times Post_t$	−2.008***	−0.428	−2.109***	−1.910***	−2.301***	−1.910***
	(0.437)	(0.852)	(0.339)	(0.448)	(0.362)	(0.438)
$Post_t$	1.068*	0.231	1.157**	1.061*	1.665***	1.021*
	(0.488)	(0.197)	(0.372)	(0.473)	(0.452)	(0.465)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Month-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Topic fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Media source fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Log likelihood	−19,375.95	−451.70	−4,223.36	−20,559.43	−3,787.43	−20,724.42
N	1,219	1,219	1,219	1,219	1,219	1,219

‡$p < 0.10$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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information diffusion process. According to Katz and Lazarsfeld (1955), information flows in two steps. New information reaches most of the public by being filtered through well-connected people who are the primary receivers of new information. In the social media setting, popular accounts are the "opinion leaders" who act as the primary vehicle through which new information is disseminated to others. Even if individuals do not trust social media platforms generally, if they trust information shared by popular accounts, those popular accounts play an influential role as a conduit of information flow. Our findings suggest that Twitter policy may prompt popular accounts to become more vigilant and cautious in passing along information. As reputation and social feedback are important motivations for participation in online social communities (Nov et al. 2010), popular accounts with established reputations are likely to be more responsive to Twitter's policy and refrain from posting or resharing misinformation, to avoid jeopardizing their credibility.

6.2.3. Active vs. Less-Active Accounts. Next, we examine heterogeneity in the effect of Twitter policy across active and less active accounts. To classify accounts into active versus less active accounts, we calculated the number of tweets per month for each Twitter account in our data set. Specifically, we divided the cumulative number of tweets posted by each account by the number of months since its Twitter join date. Then, we defined accounts in the top 10 percentile as active accounts and the rest as less active accounts. Based on this classification, we estimated the impact of Twitter policy for the active account group and the less active account group separately.

The results are presented in columns (5) and (6) in Table 6. The results indicate that the more active accounts are more responsive to Twitter's policy, as they reduce misinformation sharing more than the accounts that are less active do (Table 6, columns (5) and (6)). As positive recognition motivates users to remain active on social media (Chan et al. 2004), active Twitter users may care more about their online reputation and credibility than less active users. Consequently, when prompted with the prewarning message, active accounts are more likely to respond to the intervention.

7. Robustness Checks

7.1. Relative Time Model

As the validity of our empirical analysis depends on the assumption of parallel trends (Angrist and Pischke 2009), we estimate the relative time model (Autor 2003). Following prior literature (Autor 2003), we estimate the coefficient β separately for each time period prior to the treatment, allowing it to be different from zero before Twitter's intervention. The results of this analysis indicate that differential pretrends cannot explain the observed decline in Twitter diffusion and the number of tweets. The details of this analysis and results are presented in the Online Appendix A.[12]

7.2. Random Implementation (Shuffle) Test

We conduct random implementation (shuffle) tests to ensure that our results are not subject to false significance due to correlations in the dependent variables (Bertrand et al. 2004). Specifically, we shuffle the treatment indicator variables and randomly reassign treatments to the original data. Then, the model is re-estimated using the pseudo-treatment variables, and the random implementation process is repeated 1,000 times. Our results indicate that the distribution of the estimated coefficients is centered on zero, providing evidence that the results are not driven by the serial correlation (Online Appendix B).

7.3. Other Twitter Interventions

Twitter also implemented other interventions to curtail misinformation diffusion. If such interventions were underway during our empirical period, they might confound the effect of the intervention we focus on in this study. To check this possibility, we comprehensively searched Twitter's misinformation interventions (Table 7) where intervention 1 is the focus of our study. We found that most other interventions, interventions 3 to 15 in Table 7, were introduced on or after February 2020. Because our empirical period ended in October 2019, our results cannot be affected by them. Yet, when the "Know the Facts" policy was announced, Twitter announced that it would also no longer auto-suggest queries that may direct users to noncredible information about vaccines, which may confound the observed effect of the Know the Facts policy (intervention 2 in Table 7).[13] To alleviate the potential concern regarding this confounding factor, we conducted controlled experiments (Online Appendix I) and various robustness checks (Online Appendix H). Our controlled experiments help lessen this concern because the only manipulation in the controlled experiment is the treatment (i.e., showing Know the Facts prompt). Our robustness checks involve additional analyses after removing misinformation articles that may have been affected by the auto-suggest algorithm. Online Appendix H shows that results remain robust to removing from analysis the articles that may have been affected by the auto-suggest policy.

7.4. Additional Robustness Checks

As we noted in Section 4.3, we face a few identification challenges due to quasi-experiment setting and short lifespan of social media data. We checked the robustness of our results with several variations of model and data. First, we checked if our results are robust with a model including topic-specific time trends (Online

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Table 7. Twitter Misinformation Interventions

	Twitter misinformation intervention	Implementation date
(1)	#KnowTheFacts prompt to measles and vaccine topics	5-10-2019
(2)	Disabling auto-suggest queries that may direct to noncredible info about vaccines	5-10-2019
(3)	Expansion of #KnowTheFacts to COVID topics	1-29-2020
(4)	Announcement of their updated approach to synthetic and manipulated media (Based on the severity of content manipulation, a tweet may be labeled or removed.)	2-4-2020
(5)	Started implementing their misinformation policy announced on 2/4/2020	3-5-2020
(6)	Global expansion of the COVID-19 search prompt	3-4-2020
(7)	A dedicated COVID-19 event page	3-5-2020
(8)	COVID-19 account verification	3-20-2020
(9)	"Rethink offensive replies" beta testing (only for iOS users)	5-5-2020
(10)	Labels and warning messages to misleading COVID-19 tweets	5-11-2020
(11)	COVID-19 tab in Explore	5-18-2020
(12)	"Reply-limiting feature" beta testing	5-20-2020
(13)	"Read the article before you retweet" beta testing (only to Android users in English)	6-10-2020
(14)	"Read the article before you retweet" applies to all users	9-25-2020
(15)	Birdwatch program	1-25-2021

Appendix C). Although time-invariant heterogeneities driven by topics can be teased out by topic fixed effect, time-varying differences could also exist across topics. For example, public awareness of specific misinformation topics may change over time due to differential media coverage, government actions, and/or online platforms' product/policy changes. Such changes may further influence the public's propensity to share misinformation on a particular topic. The topic-specific time trends flexibly account for such time-varying factors by allowing unobservable topic-specific heterogeneities to change over time.

Second, we incorporate the two-step Heckman approach to account for Twitter's strategic choice of topics to be treated (Heckman 1979). In the two-step Heckman approach, the likelihood that a misinformation article to be selected to get the Know the Facts intervention is modeled separately from the total number of tweets of misinformation articles receive (i.e., *Tweet*) (Online Appendix D). Third, we use propensity score weighted estimation to additionally account for how the treatment and control groups may differ in observable characteristics (Online Appendix E). Fourth, we reconstruct the outcome variables based on the constrained 30-day time window so that Twitter activities of all articles are tracked for the same duration (Online Appendix F).

Fifth, we show that our results are robust to employing a panel analysis (Online Appendix G). The caveat of our main analysis is that we use repeated cross-sectional data instead of panel data due to the short lifespan of social media data. Although we ensured misinformation articles contain similar topics and characteristics, we acknowledge that we cannot perfectly control heterogeneities across articles and remaining heterogeneities may confound our results. Therefore, as a robustness check, we conduct a panel data analysis. Because we have information on which day each tweet is generated for each of the misinformation articles, we can aggregate the number of tweets generated for each article on a particular day. In this panel data at an article-day level, we have repeated observations of the same misinformation articles over time. Consequently, this analysis allows us to include article fixed effect that can effectively tease out inherent article-specific heterogeneities. The two-way fixed effect model with article and day fixed effects consistently shows that Twitter policy reduces the number of tweets that misinformation articles receive. Although this panel analysis has the advantage of effectively teasing out article-specific heterogeneities, only a few articles go through the diffusion process before as well as after the treatment date. Therefore, there may be limited variations in regressors over time, which can limit the use of panel analysis (Allison 2009, Hill et al. 2020). Hence, we keep this panel analysis as our robustness check. Our main results based on repeated cross-sectional data include misinformation articles spanning four years, and we incorporate several control variables and topic, time, and media source fixed effects to control for article heterogeneities.

Sixth, we conducted controlled experiments to complement our observational data analysis (Online Appendix I). This experiment addresses several endogeneity concerns. One, because Twitter strategically chose topics to apply its Know the Facts policy, systematic differences could exist across misinformation articles in treatment and control groups, and such differences may complicate the identification of effects. The results of this experiment cannot be attributed to systematic differences of misinformation articles in treatment and control groups because all participants (in both groups) are given the same set of misinformation articles. Two, this experiment helps improve internal validity. Because the

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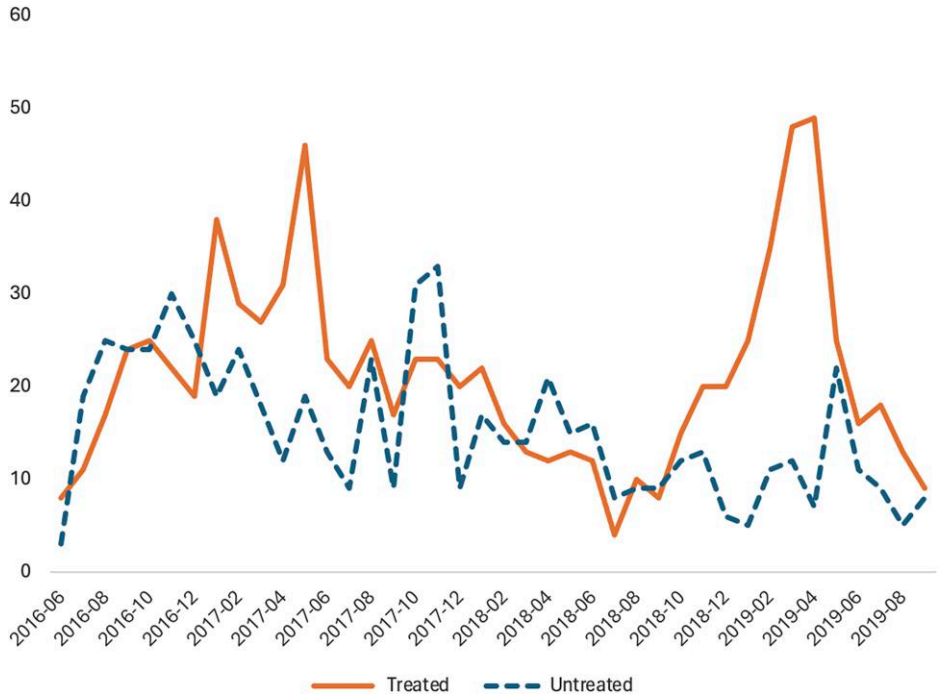
Figure 6. (Color online) Monthly Composition of Treated and Untreated Articles



only manipulation in the controlled experiment is the treatment (i.e., showing Know the Facts prompt), our findings from the experiment help us establish the direct link between Twitter treatment and behavior change of users (i.e., their likelihood of sharing misinformation). Three, because participants are randomly assigned, this experiment also alleviates the concern that the finding is confounded by user self-selection. It is reassuring that controlled experiment results are consistent with our empirical findings using observational data. Last, we examine the monthly composition of treated and untreated articles and do not find any evident compositional shift pattern (Figure 6).

8. Conclusion and Discussions

Misinformation, often dubbed "fake news," is prevalent in social media, and it spreads rapidly via social networking platforms. In this study, we explored the effectiveness of a potential cure for the infodemic we face. Specifically, we focused on the countermeasure that a popular social networking platform—Twitter—uses. Under Twitter's policy initiated in May 2019, when someone searches a topic for which misinformation is prevalent, a prompt with a link to a credible information source appears at the top of the screen, followed by tweets that include the search term. What sets Twitter's strategy apart from the more traditional tactics is its preemptive nature. Although Twitter's policy can be desirable as it does not involve censorship, its effectiveness in reducing misinformation is unclear.

Our research uncovers some promising results regarding Twitter's policy aimed at combating misinformation. Following the introduction of this policy, we observe a decrease in the prevalence of misinformation on the platform. Moreover, even when misinformation did make its way onto Twitter, such tweets were less likely to gain traction, evidenced by a decline in their being retweeted, quoted, replied to, or mentioned. This led to a decrease in the overall engagement that these erroneous tweets attracted. We delve deeper and find that the reduction is driven primarily by a decrease in diffusion by human-like tweet accounts rather than bot-like accounts. A promising result is that Twitter's policy reduces misinformation diffusion, particularly by popular accounts. Because popular accounts are often the primary culprit of disseminating misinformation on social media, the fact that Twitter's policy can alert them to become more vigilant is encouraging. Overall, our findings suggest that a misinformation policy that relies on a nudge to a credible source rather than on censorship can help contain misinformation.

Our study holds important implications for industry practitioners. Although social media platforms are not currently held legally accountable for the damage instigated by misinformation hosted on their sites, the dire societal ramifications have spurred these platforms to devise various countermeasures. The predominant strategy has been to use algorithms and human moderators to scrutinize and discard misleading content. However, this tactic has garnered criticism

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for infringing on the freedom of speech, being labeled as censorship. Our findings suggest that Twitter's nudging method, which entails subtly alerting users and guiding them toward reliable sources of information, can diminish the spread of misinformation. As this is achieved without infringing on users' freedom of speech, other platforms might want to consider the feasibility of integrating a similar strategy. However, our findings caution online platforms against relying solely on the prewarning tactic, as it cannot reduce diffusion by social bots. Last, we find that popular accounts are more responsive to the prewarning policy than less popular accounts. Considering the influential role these popular accounts play in disseminating information on social media, it might be judicious to consider broadening topics that prewarnings apply.

Our study is not without limitations. First, as noted in Section 4, our setting is a quasi-experiment where treatment is not randomly assigned, and we use repeated cross-sectional data to accommodate the transient nature of social media data. Although we included several model factors to reduce potential confounding effects, we recognize that we cannot entirely eliminate the risk that differing characteristics of misinformation articles—both before and after the treatment period and between the treatment and control groups—may introduce biases into our results. Second, it is possible that some people may not search for keywords before posting an article. In other words, the observed treatment effect is likely driven by a portion of the users instead of the entire Twitter users. Although it would be ideal to track all the activities of users before posting an article to Twitter, such data are difficult to obtain in a real-world setting. Our results that active and influential people on the Twitter platform tend to be more affected by the treatment suggest that they are more likely to search for keywords and, therefore, are exposed to the treatment of interest. Although our controlled experiment helps us establish a direct link between the treatment and the observed effect, the controlled experiment may not capture the real-world setting. We hope future researchers have access to richer real-world data that allow them to isolate the effect more cleanly and to delve into the underlying mechanisms. In addition, it is possible that Twitter's intervention could have had a spillover effect on other topics. If spillover happened to control group misinformation articles (i.e., their diffusion is also reduced), this may lead to our estimates being underestimated. Last, as we examine misinformation on a single social media platform, our results may not be generalizable to other settings. We hope future research to further examine misinformation propagation in other social media platforms and seek to further understand how users choose, consume, and propagate misinformation.

Acknowledgments

The authors contributed equally and are listed in alphabetical order. The authors thank the senior editor, associate editor, and reviewers for constructive comments and insightful suggestions.

Endnotes

[1] See https://hoaxy.osome.iu.edu/.

[2] Websites are compiled for their "50 of the biggest fake news hits on Facebook" reports.

[3] A list of websites that are identified to have posted deceptive content.

[4] A list of websites where PolitiFact has found deliberately false or fake stories.

[5] Websites that intentionally, but not necessarily solely, publish hoaxes and disinformation.

[6] See https://rapidapi.com/truthy/api/hoaxy.

[7] See https://developer.twitter.com/en/docs/twitter-api/v1/tweets/search/api-reference/premium-search.

[8] A few examples include monthly labor force surveys, television rating surveys, and political opinion polls (Steel 2008).

[9] We used the Python package TextBlob (Loria 2020) to construct the variables. The TextBlob package has been used in previous empirical research (Mousavi and Gu 2024).

[10] We also check the robustness of our results using the negative binomial model (Online Appendix J).

[11] Examples of the features include user meta-data (e.g., screenname length), information diffusion patterns (e.g., number of tweets/retweets per hour), friend statistics (e.g., number of friends distribution), temporal patterns of activity (e.g., time between two consecutive tweets), and sentiment analysis (e.g., happiness/valence/arousal scores of aggregated tweets).

[12] In the online appendix, Figure A2, we additionally estimate the relative time model without time and keyword fixed effects. We also present a plot where the distribution is winsorized at 98% and show that the estimations are robust to the winsorization.

[13] Unlike other social media platforms, including Instagram and Facebook, which have completely blocked certain search queries, Twitter took a much less proactive stance and did not block any search queries.

References

Acemoglu C, Fagnani O (2013) Opinion fluctuations and disagreement in social networks. *Math. Oper. Res.* 38:1–27.

Acemoglu O, Gheibi P (2010) Spread of (mis)information in social networks. *Games Econom. Behav.* 70:194–227.

Acemoglu D, Ozdaglar A, Siderius J (2021) A model of online misinformation. NBER Working Paper No. 28884, National Bureau of Economic Research, Cambridge, MA.

Allison PD (2009) *Fixed Effects Regression Models* (SAGE Publications, Thousand Oaks, CA).

Angrist JD, Pischke JS (2009) *Mostly Harmless Econometrics: An Empiricist's Companion*, 1st ed. (Princeton University Press, Princeton, NJ).

Autor DH (2003) Outsourcing at will: The contribution of unjust dismissal doctrine to the growth of c outsourcing. *J. Labor Econom.* 21(1):1–42.

Baltagi BH (2008) *Econometric Analysis of Panel Data*, vol. 4 (Wiley, Chichester, UK).

Bertrand M, Duflo E, Mullainathan S (2004) How much should we trust differences-in-differences estimates? *Quart. J. Econom.* 119(1):249–275.

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Cameron AC, Trivedi PK (2005) *Microeconometrics: Methods and Applications* (Cambridge University Press, Cambridge, UK).

Candogan O, Drakopoulos K (2020) Optimal signaling of content accuracy: Engagement vs. misinformation. *Oper. Res.* 68(2):497–515.

Centers for Disease Control and Prevention (2019) History of measles. Accessed June 18, 2021, https://www.cdc.gov/measles/about/history.html.

Chambers KL, Zaragoza MS (2001) Intended and unintended effects of explicit warnings on eyewitness suggestibility: Evidence from source identification tests. *Memory Cognition* 29(8):1120–1129.

Chan CML, Bhandar M, Oh LB, Chan HC (2004) Recognition and participation in a virtual community. *Proc. 37th Hawaii Internat. Conf. System Sci.* (IEEE, Piscataway, NJ), 10.

Chiou L, Tucker C (2018) Fake news and advertising on social media: A study of the anti-vaccination movement. NBER Working Paper No. 25223, National Bureau of Economic Research, Cambridge, MA.

Ecker UKH, Lewandowsky S, Tang DTW (2010) Explicit warnings reduce but do not eliminate the continued influence of misinformation. *Memory Cognition* 38(8):1087–1100.

Evrony A, Caplan A (2017) The overlooked dangers of anti-vaccination groups' social media presence. *Human Vaccine Immunotherapy* 13(6):1475–1476.

Ferrara E, Varol O, Davis C, Menczer F, Flammini A (2016) The rise of social bots. *Comm. ACM* 59(7):96–104.

Friggeri A, Adamic L, Eckles D, Cheng J (2014) Rumor cascades. *Proc. 8th Internat. AAAI Conf. Weblogs Social Media* 8(1):101–110.

Graffius SM (2022) Lifespan (Half-Life) of social media posts: Update for 2022. Accessed May 2, 2023, https:/www.ScottGraffius.com/blog/files/tag-lifespan-0028half-life0029-of-social-media-posts003a-update-for-2022.html.

Greene WH (2002) The behavior of the fixed effects estimator in nonlinear models. NYU Working Paper No. EC-02-05, New York University, New York.

Harvey D (2019) Helping you find reliable public health information on Twitter. Accessed January 5, 2021, https://blog.twitter.com/en_us/topics/company/2019/helping-you-find-reliable-public-health-information-on-twitter.html.

Heckman J (1979) Sample selection bias as a specification error. *Econometrica* 47(1):153–161.

Hill TD, Davis AP, Roos JM, French MT (2020) Limitations of fixed-effects models for panel data. *Sociol. Perspective* 63(3):357–369.

Jenco M (2019) Health officials combating measles vaccine misinformation as cases reach 1,001. Accessed June 20, 2021, https://www.aappublications.org/news/2019/06/06/measles060619.

Katz E, Lazarsfeld PF (1955) *Personal Influence: The Part Played by People in the Flow of Mass Communications* (Free Press, New York).

Kim A, Moravec PL, Dennis AR (2019) Combating fake news on social media with source ratings: The effects of user and expert reputation ratings. *J. Management Inform. Systems* 36(3):931–968.

Larson HJ, Cooper LZ, Eskola J, Katz SL, Ratzan S (2011) Addressing the vaccine confidence gap. *Lancet* 378(9790):526–535.

Loria S (2020) textblob Documentation. Release 0.16. Accessed May 2, 2023, https://buildmedia.readthedocs.org/media/pdf/textblob/latest/textblob.pdf.

Moravec PL, Kim A, Dennis AR (2020) Appealing to sense and sensibility: System 1 and system 2 interventions for fake news on social media. *Inform. Systems Res.* 31(3):987–1006.

Mostagir M, Siderius J (2023) Social inequality and the spread of misinformation. *Management Sci.* 69(2):968–995.

Mostagir M, Ozdaglar A, Siderius J (2022) When is society susceptible to manipulation? *Management Sci.* 68(10):7153–7175.

Mousavi R, Gu B (2024) Resilience messaging: The effect of governors' social media communications on community compliance during a public health crisis. *Inform. Systems Res.*, ePub ahead of print July 20, 2023, https://doi.org/10.1287/isre.2021.0599.

Nov O, Naaman M, Ye C (2010) Analysis of participation in an online photo-sharing community: A multidimensional perspective. *J. Amer. Soc. Inform. Sci. Tech.* 61(3):555–566.

Novacic I (2020) Censorship on social media? It's not what you think. Accessed June 18, 2021, https://www.cbsnews.com/news/censorship-social-media-conservative-liberal-cbsn-originals/.

Petrovic M, Roberts R, Ramsay M (2001) Second dose of measles, mumps, and rubella vaccine: Questionnaire survey of health professionals. *Primary Care* 322:4.

Schul (1993) When warning succeeds: The effect of warning on success in ignoring invalid information. *J. Experiment. Soc. Psych.* 29:42–62.

Seetharaman D (2016) Facebook bans fake news sites from using its advertising network. Accessed July 26, 2021, https://www.wsj.com/articles/facebook-bans-fake-news-sites-from-using-its-advertising-network-1479175778.

Shao C, Ciampaglia GL, Flammini A, Menczer F (2016) Hoaxy: A platform for tracking online misinformation. *Proc. 25th Internat. Conf. Companion World Wide Web* (ACM, New York), 745–750.

Shao C, Ciampaglia GL, Varol O, Yang KC, Flammini A, Menczer F (2018) The spread of low-credibility content by social bots. *Nature Comm.* 9(1):4787.

Steel (2008) Repeated cross-sectional design. Lavrakas PJ, ed. *Encyclopedia of Survey Research Methods* (Sage Publications, Thousand Oaks, CA), 715–716.

Stella M, Ferrara E, De Domenico M (2018) Bots increase exposure to negative and inflammatory content in online social systems. *Proc. Natl. Acad. Sci. USA* 115(49):12435–12440.

Suciu P (2019) More Americans are getting their news from social media. Accessed June 18, 2021, https://www.forbes.com/sites/petersuciu/2019/10/11/more-americans-are-getting-their-news-from-social-media/.

Varol O, Ferrara E, Davis CA, Menczer F, Flammini A (2017) Online human-bot interactions: Detection, estimation, and characterization. *Proc. 11th Internat. AAAI Conf. Web and Social Media* (AAAI Press, Palo Alto, CA), 10.

Vosoughi S, Roy D, Aral S (2018) The spread of true and false news online. *Science* 359(6380):1146–1151.

Wardle F (2018) Fake news. It's complicated. Accessed June 18, 2021, https://medium.com/1st-draft/fake-news-its-complicated-d0f773766c79.

Yang K, Varol O, Davis CA, Ferrara E, Flammini A, Menczer F (2019) Arming the public with artificial intelligence to counter social bots. *Human Behav. Emerging Tech.* 1(1):48–61.